

Newton's Laws of Motion



2. Net Force

NEWTON'S LAWS OF MOTION

2. $F = m \cdot a$

“NET FORCE IS EQUAL TO MASS TIMES ACCELERATION”

The diagram illustrates the equation $F = m \cdot a$ using color-coded boxes and labels. On the left, a purple box contains the vector $\vec{F}$ with the word "force" above it, and a pink box contains the subscript n with the word "net" below it. In the center is an equals sign. On the right, a blue box contains the variable m with the word "mass" above it, and a yellow box contains the vector $\vec{a}$ with the word "acceleration" below it.

$$\begin{array}{c} \text{force} \\ \vec{F} \end{array} \begin{array}{c} n \\ \text{net} \end{array} = \begin{array}{c} \text{mass} \\ m \end{array} \begin{array}{c} \vec{a} \\ \text{acceleration} \end{array}$$

Quantity	Definition	Example	Formula
Acceleration	Vector quantity showing the change in velocity	The car came to a stop in 1 second. -11 m/s²	$a = \frac{\Delta v}{\Delta t}$ $a = \frac{\text{change in velocity}}{\text{change in time}}$ $a = \frac{v_f - v_0}{t_f - t_0}$
Mass	Scalar quantity showing how much mass, or inertia, an object has.	The car has a mass of 1,500 kg .	measured in kg.
Force	Vector quantity showing the intensity of a push or pull	The tow truck pulled the car with a force of 3,000 N .	$F_n = m \cdot a$ Net Force = mass · acceleration $F = m \cdot \frac{v_f - v_0}{t_f - t_0}$

Practice

A box with a mass of 50 kg is pushed with a force of 100 N to the right.

There is no friction or air resistance acting on the box.

Using the GUESS method, find the acceleration of the box.

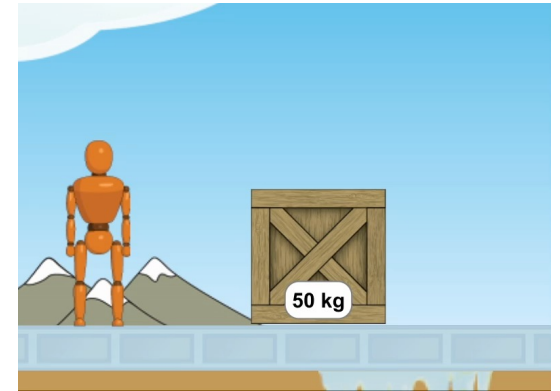
G $F_1 = 100 \text{ N right.}$ $F_f = 0 \text{ N left}$
 $m = 50 \text{ kg.}$

U $a = ?$

E $F_n = m \cdot a \rightarrow \frac{F_n}{m} = a \rightarrow a = \frac{F_n}{m} \rightarrow a = \frac{(F_1 + F_f)}{m}$

S $a = \frac{100 \text{ N} + (-0 \text{ N})}{50 \text{ kg}}$

S $a = \frac{100 \text{ N}}{50 \text{ kg}} \rightarrow a = 2 \frac{\text{kg} \cdot \frac{\text{m}}{\text{s}^2}}{\text{kg}} \rightarrow a = 2 \frac{\text{m}}{\text{s}^2} \text{ right}$



Click to
Check!

Simulation Questions

Go to:

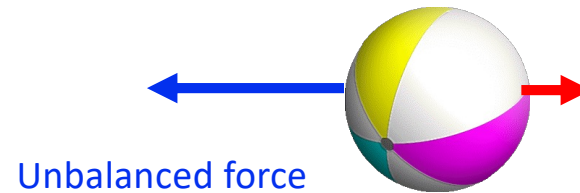
https://phet.colorado.edu/sims/html/forces-and-motion-basics/latest/forces-and-motion-basics_all.html

1. Click the "acceleration" option.
2. Check all the information boxes.
3. Set the mass to 50 kg.
4. Apply 50 N of force.
What happened? Why?
5. Set friction to "none"
What is the acceleration now?
What happened to the speed?
6. Apply 50 N of force in the opposite direction.
What happened to the acceleration?
What happened to the speed?
7. Apply 100 N of force in the same direction as step 6.
What happened to the acceleration?
What happened to the speed?
8. Add friction.
What happened to the acceleration?
What happened to the speed?
9. Try to get a constant speed of 20 m/s.
How did you do it?
10. Double the mass.
What happens?

Challenge

A ball with a mass of 0.5 kg is hit with a force of 1N to the left. The total friction resisting the force is 0.2 N right to the right. Using the GUESS method, find the acceleration of the ball.

G $\vec{A} = 1\text{N left}$ $m = 0.5 \text{ kg}$
 $\vec{f} = 0.2\text{N right}$



U $a = ?$

E $F_n = m \cdot a \rightarrow \frac{F_n}{m} = a \rightarrow a = \frac{F_n}{m} \rightarrow a = \frac{(\vec{A} + \vec{f})}{m}$

S $a = \frac{1.0\text{N} + (-0.2)\text{N}}{0.5\text{kg}}$

S $a = \frac{0.8\text{N}}{0.5\text{kg}} \rightarrow a = 16 \frac{\text{kg} \cdot \frac{\text{m}}{\text{s}^2}}{\text{kg}} \rightarrow a = 16 \frac{\text{m}}{\text{s}^2}$